

Digital Logic

BCA — Semester I — 2024 — Answer Sheet
Subject Code: CACS103

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Define Digital computer. Subtract: $(101010.101 - 1000100.001)$ using both 1's and 2's complement.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What do you mean by K-map? Explain the K-map with three variables. Simplify $(F(p,q,r,s) = \sum(3,4,7,8,14))$ with don't care conditions $(d(p,q,r,s) = \sum(1,6,9,13))$. Design the logic circuit using minimum NAND gates.]

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Differentiate between combinational and sequential logic circuits. Implement a full adder circuit using a decoder and two OR gates.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define priority encoder. Explain 8-to-3 priority encoder in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain the duality theorem with an example. Draw logic gates that implement the expression:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. How does a flip-flop differ from a latch? Explain clocked SR flip-flop with logic diagram, truth table, characteristic table, and excitation table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write short notes on (any two): a) State reduction table b) Multiplexer c) Synchronous and Asynchronous counter

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Differentiate between PAL and PLA. Design a combinational circuit with four inputs (BCD digit) and four outputs lines that generate the 2's Complements of the binary input pattern with circuit diagram, truth table, and block diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain shift register with parallel load. Design a synchronous Mod-10 counter to count in the sequence 0,2,4,5,6,8 using T flip-flops.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Explain how race condition in JK flip-flop can be resolved. A sequential circuit with two D Flip-Flops (A, B), inputs (x, y) , and output (z) is defined by:

$$\begin{aligned} A(t+1) &= xy' + xB \\ B(t+1) &= x'B + xA \\ z &= A \end{aligned}$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.